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New 3D printed metal alloy enhances durability for space exploration

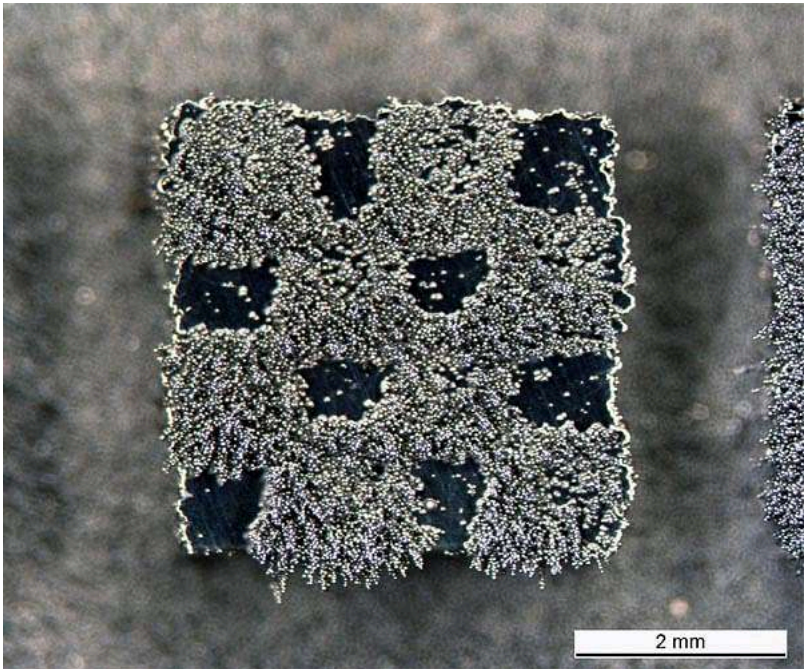


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New 3D printed metal alloy enhances durability for space exploration

by Riko Seibo

Tokyo, Japan (SPX) Nov 01, 2024

The research team at Korea Institute of Materials Science (KIMS), led by Dr. Jeong Min Park of the Nano Materials Research Division, has developed a highly durable metal 3D-printed alloy designed to meet the demands of space exploration. Collaborating with Professor Jung Gi Kim of Gyeongsang National University and Professor Hyoung Seop Kim of POSTECH, the team created an alloy optimized for the extreme cryogenic temperatures often encountered in space, achieving notable mechanical performance down to -196 C.

The researchers enhanced a CoCrFeMnNi alloy by introducing a small amount of carbon, which significantly improved the alloy's properties in cryogenic conditions. Using the Laser Powder Bed Fusion (LPBF) method, a form of metal additive manufacturing, they achieved a fine distribution of nano-carbides along the nano-sized cellular boundaries within the alloy. This approach strengthened the alloy by over 140% in both tensile strength and ductility compared to similar carbon-free alloys in cryogenic conditions. Impressively, the alloy's elongation was twice as high at 77 K as it was at 298 K. These results suggest new guidelines for alloy design in additive manufacturing, offering a roadmap to produce load-bearing components for use in extreme, low-temperature environments.

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This advanced 3D printing method enables precise microstructural control, making it especially suitable for intricate components like fuel injectors and turbine nozzles in space exploration rockets. The new alloy extends the lifespan and boosts the performance of these parts by overcoming the traditional limitations of low-temperature toughness seen in standard 3D-printed alloys.

Dr. Jeong Min Park, senior researcher and project leader, noted, "This research presents a significant breakthrough in developing new alloys for extreme environments, offering new possibilities. Through 3D printing technology that surpasses the manufacturing limits of conventional space exploration components, we can significantly improve the performance of parts used in space launch vehicles."

Funded by KIMS' foundational projects, including "Development of design for additive manufacturing to develop superhard heterogeneous materials with complex design" and "Development of High Performance Materials and Processes for Metal 3D Printing," this research was published in the prestigious journal *Additive Manufacturing* (Impact Factor: 11). Looking forward, the research team aims to explore the commercialization potential of this technology while conducting further studies to confirm its robustness in challenging environments.

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